Conduction in RealQM:
Free Boundary as Carrier

Claes Johnson[1]

[1]KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,
claesjohnson@gmail.com

August 27, 2026

Mathematical formulation by the author; analysis developed with AI assistance (Claude, Anthropic).

Abstract

RealQM models matter as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy. This article asks whether such a construction **conducts**, and finds the answer to turn on two things: what is carried, and what kind of conductor is being asked for.

The second is settled first, because the construction does not have a free choice. Localising one electron per domain makes it the Hubbard model at $U \to \infty$ — a Mott insulator at every density, as the barrier scans below find by direct computation. Metals are therefore *out of scope*, in kind and not by degree: their carriers are extended Bloch states and their defining measurements are properties of a degenerate electron gas. What is within scope is matter whose carrier is a localised charge on a site, whose gap is on-site repulsion rather than a band structure, and whose transport is activated hopping — doped Mott insulators, organic and molecular semiconductors, polaronic oxides, amorphous semiconductors. Silicon and the III–Vs are excluded with the metals, their gaps being band gaps. Two results usually read as failures follow from the scope rather than against it: the prediction $d\sigma/dT > 0$ is *correct* for an activated conductor and fatal only for copper; and the absence of a Fermi surface, which no arrangement of densities on domains appears able to supply, does not bear on a dilute carrier gas, which is non-degenerate and has none to reproduce.

The scope is nonetheless a class the construction belongs to rather than a set of materials it describes, and the reason is measured here. On a ring validated against the sliding calculation to five decimals, a neutral chain with one site ionised gives a carrier that localises unaided in the interstitial pockets, with an activation energy of 4.9 eV at $a = 2.2\,a_0$ — and 9.0 eV when the lattice is compressed to $1.4\,a_0$. Compression *doubles* the barrier rather than closing it, so there is no Mott transition and no metallic state at higher density, exactly as the $U \to \infty$ identification predicts: compression cannot increase overlap when non-overlap is the postulate. Against the 10–500 meV of a real hopping semiconductor that is ten to a hundred times too large. *As formulated the framework conducts nowhere*, and the repair identified below — occupation two at a computed on-site cost, finite U in place of infinite — becomes the only route to any conduction rather than one option among several. The same runs withdraw an earlier conclusion: the corona result, an added electron delocalising with a flat energy, carried a net charge, and at neutrality the carrier localises instead.

Not an electron. An electron here is a unit of charge density on a domain; what exists is the *partition* of space, and transport is the partition pattern travelling. What moves is the **free boundary** — a locus where two densities balance — in the relation a dislocation bears to the

slip it produces: as a kink passes, each electron advances one lattice spacing and stops while the kink crosses the whole chain, so charge genuinely arrives at the far end while no constituent moves far. Nothing material traverses the sample. That diagnosis also accounts for a series of null results reported here, each of which imposed a carrier obliged to translate bodily through occupied space, and so measured the price of forbidding the only mechanism available: with the free interface, a uniform density is admissible on any partition, redrawing a boundary costs nothing, and the boundary is the only object the framework moves.

Numerically, an extra electron on a ring of hydrogens does not sit anywhere: it relaxes to a uniform corona, at $a = 2.2\,a_0$ and still at $a = 4.0\,a_0$ where a real electron would occupy one atom as H^-, so the framework over-delocalises and never localises charge. A full electron ring against bare kernels is by contrast **pinned**, rotating against the ion ring at a cost of 0.01186 Ha = 323 meV for six electrons — 54 meV each, a threshold field of 9×10^8 V/m, or 0.11 V per lattice site against the 4×10^{-12} V per site of copper carrying an ampere. Rigid sliding is therefore excluded as a conduction mechanism by ten orders of magnitude. It is however an upper bound, since it obliges every electron to crest its barrier at once; the cheap path is a kink, whose cost is independent of chain length; the elementary defect from which one is built is measured below at 179 meV and is itself length-independent to 5%, but the kink energy proper is not reported here. Whether the two ring results additionally constitute a semiconductor — filled complement pinned, added carrier free — is left as a conjecture, the two runs differing in screening as well as in filling.

The same free interface that supplies the mechanism damages the equation of state, and the two cannot be repaired independently. Partitioning a gas of density n gives a kinetic energy density $t(n) = C\,n^{5/3}$ — the Thomas–Fermi form from partition geometry rather than Fermi statistics — but the flat trial function is admissible, so $C = 0$ exactly: no degeneracy pressure, a *vanishing* screening length since $k^2 \propto 1/C$ (the framework over-screens rather than failing to screen), a dispersionless plasmon, no bandwidth. The defect is nonetheless one number and not the construction, C being fixed by the interface condition: Robin at $\beta a = 1.146$ gives $C = 2.871 = C_{\mathrm{TF}}$ exactly against Dirichlet's 14.804, all three carrying the correct $n^{5/3}$ scaling. Whether that value is tolerable for atoms and molecules is the open question, and it is sharper than a parameter check — Thomas–Fermi binds no molecule at all, by Teller's theorem, so the far end of the dial is hostile to exactly what this framework does best.

Two further results are recorded. The alkali series Li–Cs is reproduced to a mean error of 6.2% against the standard treatment's 5.6%, carrying neither exchange nor band filling, but this is shown *not* to test the quantum treatment, since setting the electronic kinetic energy to zero altogether moves the radii by one per cent: the lattice scale of a simple metal is electrostatic. And the high-temperature limit is the *easier* case for this framework rather than the harder one, degeneracy not being required there — $\omega_p^2 = 4\pi n$ comes out exactly, for the structural reason that a rigid translation leaves the localisation term untouched. What remains genuinely open is the Fermi surface, which no arrangement of densities on domains appears able to supply.

A last result is methodological, and on reflection it is the load-bearing one, since every number above that concerns transport depends on it. The free boundary is evolved by *local* rules: a cell changes hands when the neighbouring density exceeds its own. A run therefore descends to the nearest configuration no single such change improves — a stuck point, not a minimum — and which one it reaches depends on where it started. Four independent geometries were measured and all four agree: energies that ought to be properties of the system spread by 0.005–0.05 Ha across seeds, across builds, or fail to settle at all, against the 0.001–0.02 Ha barriers being sought. The ambiguity exceeds the quantity by one to two orders. It is unrepaired by the surface tension β, by reducing the boundary step, by freezing the labels, and by changing to a test system without the pathologies of the first. It is a failure of the *search* and not of the energy functional, which is why nothing in the functional touches it, and it is why the ring's rigid-sliding figure moved under a recompilation that altered no physics.

The repair is not a better minimiser but none at all. The domain walls of a chain are planes, so they can be *placed*: specify the partition, freeze it, relax only the wavefunctions, and ask what a stated configuration costs rather than where a boundary goes. Repeated runs then agree to all five decimals recorded, which no free-boundary quantity here does. On that footing a misplaced domain wall costs 179 meV and the figure is independent of chain length to 5%; the mismatch of a chain carrying a vacancy gains 0.112 Ha by delocalising, again length-independent; and the barrier to sliding a pattern through its lattice is tunable through zero by density and by core radius. The quantity that governs conduction — the activation energy for a kink to *move* — is not among them: three constructions were attempted and each was rejected by an invariant it failed, the last because translating a kink in a short chain measures the vacancy's preference for the chain's centre and not the lattice's grip on it. It is left open with a stated route rather than estimated, which is the honest position and is the reason this article is about the carrier and the difficulty of costing it rather than about a conductivity.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system, and the question pursued here is the elementary one such a system poses — whether it *conducts*, and if so what carries the charge. The same construction covers a stellar interior and the recombination era of a Coulomb cosmology, which are treated as the high-temperature limit in §7.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \tag{1}$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

A two-component system of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [4] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 What kind of conductor this can be

It is worth settling the target before the measurements, because the framework does not have a free choice of one and the wrong choice makes its central results read as failures.

The construction localises charge by design: one electron, one domain, non-overlapping. As §3 sets out, that is the Hubbard model in the limit $U \to \infty$ — double occupancy forbidden, every site singly occupied — which is a Mott insulator at every density, and which the barrier scans found by direct computation before the identification was made. A metal is the opposite object. Its carriers are extended Bloch states spanning the crystal, its conduction is band motion, and its defining measurements — de Haas–van Alphen oscillations, a linear electronic specific heat, positive plasmon dispersion — are properties of a degenerate electron gas with a sharp Fermi surface. None of that is available here, and this article does not claim it.

What is within reach. Matter in which the carrier is a *localised* charge on a site, the gap is on-site repulsion rather than a band structure, and transport is thermally activated hopping between sites:

doped **Mott insulators**	the gap is U; the framework is the $U \to \infty$ limit of their model
organic and molecular semiconductors	hopping is the accepted description, not an approximation
polaronic oxides	the carrier is a self-trapped charge, which is what a domain is
amorphous semiconductors	variable-range hopping, no bands anywhere in the account

What is not. Silicon, germanium and the III–Vs have *band* gaps — gaps opened by a periodic potential acting on delocalised states — together with effective masses and band transport. A theory without bands cannot supply those, and questions of mobility against effective mass stay closed to it. The distinction is not a matter of degree: a Mott gap and a band gap have different origins, and only the first is available here.

The right class, but not the right regime. One qualification belongs with this scope before anything is claimed for it, and it is quantitative. Activated transport by localised carriers is indeed what this construction does. But a hopping semiconductor has an activation energy of some 10–500 meV, and the barrier measured in §3 is 4.9 eV at $a = 2.2\,a_0$, rising to 9.0 eV under compression — ten to a hundred times too large, and worsening in the direction that should help. Those are rigid-lattice figures and therefore upper bounds, the polaron correction being unmeasured (§3); but the gap to be closed is one to two orders of magnitude, and lattice relaxation is not usually worth more than a factor of a few. A material with a 9 eV activation energy is not a semiconductor but an insulator with no accessible transport at any temperature. *As formulated the framework does not conduct* — though see §3, where a carrier prevented from re-binding to its donor lowers the barrier substantially, though the cost per carrier rises again as the doping is thinned towards realistic values. What follows is therefore the statement of a class the construction belongs to, not a claim that it describes the materials in it; and the repair of §3, finite U in place of infinite, ceases to be one option among several and becomes the only route to any conduction at all.

Two consequences, and they run opposite to how these results have been read. First, the prediction $d\sigma/dT > 0$ derived in §3 is *correct* for this target rather than fatal to it. Conductivity rising with temperature is what an activated hopping conductor does; it is only fatal if the target is copper. Second, and more consequentially, the Fermi surface ceases to be an unresolved failure and becomes a scope limit. A Fermi surface is a property of a degenerate electron gas; a dilute carrier gas is non-degenerate, Boltzmann rather than Fermi statistics govern it, and there is no E_F to reproduce. *The framework's largest lacuna does not bear on the material class it can describe.* §8 states what that costs if one wants metals after all.

3 Conduction

Electric conduction is the interaction of the electrons with the ion lattice. It is the elementary case — ordinary matter at ordinary temperature — and it is for that reason the *demanding* one here, since ordinary-temperature conduction is degenerate matter, and degeneracy is what this framework has least of.

One property of the construction is needed before anything else, and it is the property that makes conduction a live question rather than a hopeless one. With the free interface used throughout, a uniform density is admissible on any partition: $\psi_i \equiv |\Omega_i|^{-1/2}$ satisfies the normalisation, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Redrawing the boundary therefore costs nothing,

$$\text{a free interface between domains of equal density carries no confinement energy,} \tag{2}$$

and the boundary is the only object this framework moves — its solver evolves each interface by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, transferring ownership of a cell when the neighbour's density exceeds its own. Domains do not translate; they are redrawn. *A vanishing interface cost is a vanishing cost of transport.* What that same fact does to the equation of state is a separate matter and is taken up in §6, where it is much less welcome.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$. Within the scope set in §2 that is the *right* answer, activated transport being what a localised-carrier semiconductor does; it would be fatal only for a metal, which is why the target matters. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that carried the metals in §4 is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $HF + H_2O \to F^- + H_3O^+$ and $HCl + NH_3 \to Cl^- + NH_4^+$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

Which filling the measurements belong to, and which needs the repair. It is worth being exact about this, because it determines how much of the article depends on β. At *half filling* — one electron per site, every domain occupied — the $U \to \infty$ construction forbids conduction outright: there is no vacancy for a carrier to enter, and the insulator is structural rather than dynamical. That is the case the double-occupancy repair is for. But it is not the case measured here. The chains reported above are doped by ionising every second atom, leaving half the sites empty, and every barrier in this article is the cost of a carrier moving into a *vacancy.* Single occupancy handles that without amendment.

So the transport results stand on their own: they require no finite U, no double occupancy and no β. What requires the repair is the undoped, half-filled chain, which is not run here. This is the stronger position of the two — the measurements are independent of the proposal made to extend them — and it locates precisely what the proposal would buy: not better hopping, but conduction at a filling where the construction currently permits none.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

What the framework actually does with an extra electron. The clean test is a ring, which has no ends and in which every site is equivalent by construction, so the energy must repeat with the lattice period — a validity check the open-chain scans lacked. Six hydrogens on a ring, one extra unit of charge in the corona outside them, slid round by angle:

arc spacing	over an atom	quarter way	between two atoms
$a = 2.2\,a_0$	-3.51	-3.51	-3.51
$a = 4.0\,a_0$	-3.30	—	-3.30

The energy is flat to the resolution available, and the reason is visible in the solution rather than inferred: the extra charge does not sit anywhere. It relaxes to a **uniform corona around the whole ring**, indifferent to where it was placed. There is no localised carrier to move, so there is nothing to corrugate the path and no resistance to measure.

That holds at $a = 4.0\,a_0$, nearly three times the H_2 bond length, where a real extra electron would occupy a single atom as H^-. *The framework therefore over-delocalises: it spreads charge at every separation tested and never localises it.*

The collective mode, which is how a metal actually conducts. A single carrier is the wrong object. Conduction is not one electron moving past stationary others but the whole distribution shifting while the ions stay, and the corresponding energy is the **pinning** of that distribution to the lattice — the quantity that decides whether a charge-density wave slides or sticks, and the Frenkel–Kontorova barrier for friction. On a ring it is directly computable: put n bare ions on an inner ring and n electron domains on an outer one, rotate the electron ring by ϕ site spacings, and record the energy.

The construction carries its own check, and passes it exactly: a rotation by one full spacing maps each electron onto its neighbour's place, so $E(1)$ must equal $E(0)$, and it does to five decimals. That also establishes that the solver is *deterministic*, so differences between configurations are real rather than scatter.

$\phi = 0$, electrons over ions	-3.36796
$\phi = 0.5$, electrons between ions	-3.35610
pinning	0.01186 Ha $= 323$ meV $= 12.5\,k_BT$

The sea is **pinned**, with the electrons preferring to sit over the ions. That it is pinned at all is worth pausing on, because §6 finds that a free interface should screen perfectly: a sea that redistributes at no kinetic cost should flatten the ion potential and pin to nothing. What prevents it is that the charge here cannot redistribute continuously — it is quantised into six indivisible units, one to a domain. The framework's own signature postulate is therefore what blocks the screening its kinetic functional would otherwise supply, which is once more the tension recorded below: the rule that explains why charge comes in units is the rule that forbids a metal. Six electrons give $12.5\,k_BT$, so about $2\,k_BT$ each, and a collective barrier scales with the number of participants.

What the pinning means as a field. Six electrons give $12.5\,k_BT$, so 54 meV each, peak to peak, across half a site spacing. Barrier and field-work both scale with the number of participants, so the *threshold field* for depinning is intensive, and therefore transferable off this small ring: taking the corrugation as sinusoidal, $E_T = \pi\,\Delta E/(NeL) = 9 \times 10^8$ V/m.

threshold field, rigid sliding	9×10^8 V/m
potential drop per lattice site	0.11 V
across a centimetre of wire	$\approx 10^7$ V
copper carrying 1 A, per site	4×10^{-12} V

Ten orders of magnitude. *Rigid sliding is not conduction*, and that is now a quantitative exclusion rather than an impression.

The kink, and why a chain rather than a ring. Rigid sliding is nevertheless an *upper bound* on the cost of transport, since it obliges every electron to crest its barrier at once and its cost grows with the number of participants. The cheap path is a **kink**: advance the electrons on one side of a point by one full spacing over some width, leaving one ion unpartnered. Only the kink region pays, so the cost is independent of the chain's length, and one kink crossing the chain transports exactly one electron. This is how a charge-density wave actually depins, and it is the soliton form of hopping.

A kink is native to a chain rather than to a ring, for a reason that also bounds the ring result. No electrostatic bias can drive a sea around a closed loop: a conservative field has zero circulation. The ring supplies a clean *barrier* — no ends, every site equivalent, the energy obliged to repeat with the lattice period — but it cannot host the *drive*. A chain has ends, a potential drop along it is an ordinary thing to apply, and a single kink may enter at one end and leave at the other, which is the transport event itself. On a ring kinks must come in pairs.

Two numbers settle it: the creation energy $E(\text{kink}) - E(\text{uniform})$, whether a kink exists at all; and the migration barrier, the variation of the energy as the kink is stepped along, which is the Peierls–Nabarro barrier and fixes the threshold field for kink motion. If migration falls far below the 54 meV of the rigid mode, the chain conducts at fields the rigid mode could never reach. The creation energy is now computed and is reported below; the migration barrier is not, and remains the open half of the pair.

The partition must be specified, not searched for. Before the number, the method, because the obvious approach fails and the failure is instructive. A kink is *not* the ground state, so a free-boundary minimisation relaxes it away — asked to hold one, the solver oscillates between the kinked and uniform partitions and the energy never settles, wandering over 0.05 Ha against a barrier of order 0.01. That is not a convergence failure to be tuned away: it is unaltered by the surface tension β, by reducing the boundary step fivefold, and by freezing the labels outright.

The cause is general and it disqualifies the free boundary for every transport quantity in this article. The interface moves by *local single-cell label flips*, so a run descends to whichever configuration no single flip improves — the nearest stuck point, not the minimum — and which one that is depends on the initial condition. Four independent geometries agree:

geometry	spread	against a barrier of
ring, between two builds	0.005 Ha	0.012–0.022
kink chain, across steps	0.05 Ha (never settles)	~ 0.01
$H^- + H$, four seeds	0.031 Ha	~ 0.001
alkali chain, two seeds	0.013 Ha	~ 0.001

In every case the ambiguity exceeds the quantity sought, by one to two orders. It is a failure of the *search*, not of the energy functional, which is why nothing in the functional touches it. It is also why the ring's 0.01186 Ha moved to 0.022 Ha under a recompilation that altered no physics: rounding at the last bit flips cells sitting near the threshold, and a different stuck point follows. The rigid-sliding figure should be read as a bound of that order rather than a determination, and the doping barriers of Table 1, obtained the same way, with it.

The repair is not a better minimiser but no minimiser. The domain walls of a chain are planes perpendicular to the axis, so they can be *placed*: specify the partition, freeze it, and relax only the wavefunctions. Nothing is searched for, the same input returns the same output, and the question asked is no longer "where does the boundary go" but "what does this configuration cost" — which is the question a barrier actually poses. It is the device that makes the H_2 midplane exact, applied to a coordinate instead of a symmetry.

The cost of a misplaced domain wall, and that it does not grow. What follows is *not* the kink of the preceding paragraph, and the distinction matters. A kink is a discommensuration: the registry slips by one full spacing across it, one ion is left unpartnered, and the slip is spread over a width. That requires the number of domains and the number of cores to differ locally — N domains in N wells is perfectly commensurate and admits no kink at all. What is computed here is the elementary defect a kink is built from: a *single misplaced domain wall*, sitting on a core instead of midway between two, sharp and one site wide, with the registry unchanged on either side. It is an upper bound on nothing in particular and a lower bound on nothing either; it is a well-defined local defect energy, and it is reported because it is reproducible where the free-boundary numbers are not.

Lithium-like cores, one valence electron each, a pseudopotential radius $r_c = 1.6$ and spacing $a = 6.0$ a_0. The uniform configuration puts every wall midway between neighbouring cores; the defect displaces the *middle* wall by $a/2$ and leaves the rest, so one domain widens and its neighbour narrows while everything beyond is untouched — and the end domains, identical in both, cancel exactly in the difference:

cores	uniform (Ha)	kinked (Ha)	$E(\text{kink}) - E(\text{uniform})$
4	-1.03860	-1.03220	0.00640 Ha = 174 meV
5	-1.29985	-1.29310	0.00675 Ha = 184 meV

The defect costs ≈ 0.0066 Ha $= 179$ meV $\approx 7\,k_BT$, and the cost does not grow with the chain. Adding a whole site changes it by 0.00035 Ha while the chain's total energy changes by 0.26 Ha — a factor of 750. That is the property the mechanism turns on: rigid sliding costs 54 meV per electron and obliges all of them at once, so its activation is $\sim 5 \times 54 = 270$ meV at five sites and 5.4 eV at a hundred, while the kink stays at 179 meV however long the chain. At these short lengths the two are already comparable and the kink is lower; at any physical length the comparison is not close.

Two caveats belong with the figure, and a third with its interpretation. Chains of four and five sites are short, so the length-independence is demonstrated over a narrow range even though it holds to 5% there. The defect profile is the sharpest possible, one wall wide. And the step from this quantity to the kink energy proper is not taken here: a kink needs N domains over $N \pm 1$ cores, which an equal-count chain cannot host, so it must be built either on a chain with a vacancy or as a kink–antikink pair with the domain counts locally compressed and dilated. A pair constructed instead by displacing the walls rigidly by one spacing was tried and rejected: it collapses the domain at the antikink to zero width. That construction, and the migration barrier with it, remain the open half of the pair of numbers named above.

A third result falls out of the same scan and confirms an earlier one by a different route. Sliding all the walls together, the *bond*-centred registry lies 0.064 Ha below the atom-centred one (-1.364 against -1.29985): the electrons prefer to sit between two cores rather than on one. That is the interstitial-pocket preference the compression test finds below, obtained here without a free boundary anywhere in the calculation.

The compression test, and the absence of a Mott transition. The barrier scans of Table 1 were open to the objection that they impose a localised carrier and so measure the cost of a constraint. That objection can now be removed. On a ring — validated by reproducing the sliding calculation above to five decimals, -3.368 — a *neutral* chain of six sites with one site ionised, its electron placed in a shell, gives a carrier that localises of its own accord in the interstitial pockets, the on-site positions being the maxima. Sliding it round gives the activation energy directly, and compressing the lattice tests for a Mott transition:

spacing	pocket (Ha)	saddle (Ha)	barrier
$a = 2.2\,a_0$	-3.264	-3.106	0.180 Ha = 4.9 eV
$a = 1.4\,a_0$	-3.230	-2.900	0.330 Ha = 9.0 eV

Compression nearly doubles the barrier. A factor 1.57 in density raises it by 1.8, in the direction opposite to a metal–insulator transition, so there is no metallic state waiting at higher density. This is what §3 predicts and had not measured: compression cannot increase overlap when non-overlap is the postulate, so it only shrinks each domain, and no finite U can be beaten by a bandwidth that does not exist. It also settles the standing doubt about Table 1, whose trend it reproduces on a validated geometry with an unconstrained carrier.

A second consequence bears on the corona result. That run — an extra electron delocalising to a uniform shell with a flat energy — carried a *net charge*, seven electrons against six protons, and a charged cluster in vacuum spreads for reasons unconnected with conduction. The neutral version of the same experiment, reported here, does the opposite: the carrier localises, with a barrier of several electron-volts. The conclusion drawn from the corona, that the framework never localises charge at any separation, does not survive at neutrality and is withdrawn.

A carrier that cannot re-bind, and a barrier five times smaller. Every figure above shares an assumption that turns out to matter more than the geometry. At zero temperature an ionised donor beside a free carrier is not a ground state — it is what temperature or a band buys — so an energy minimisation will put the carrier back onto its donor if it can, and a calculation intended to measure transport instead measures binding. Several electron-volts is what climbing out of a donor well costs; sliding a screened carrier along a channel should not be.

The remedy is core–valence separation imposed by construction, which is what a pseudopotential does: hold the carriers outside a cylinder about the chain axis, enforced at every step rather than merely seeded, and leave the chain's own electrons free to screen the bare protons as they would otherwise. On a straight chain of seven hydrogens with every second one ionised — three carriers in an outer sheath, neutral overall — sliding the sheath along the chain gives

	ring, carrier free	straight chain, carriers held outside $r = 1.0\,a_0$
carriers	1	3
total barrier	0.180 Ha	0.033 Ha
per carrier	**4.9 eV**	**0.30 eV**

The constrained figures are -3.560 at the sites and -3.527 midway, against a floor of -4.11 Ha for seven hydrogens, so they are physical; and the chain being symmetric about its centre atom, $E(-d)$ must equal $E(d)$, which it does to 0.002 Ha against a total of 0.033.

The division by carrier count is not legitimate, and the dilute limit is the stiff one. That reading assumed the carriers to be independent, which a change of doping falsifies. Removing one donor from the same chain — three carriers to two, 43% to 29%, everything else held — more than doubles the total barrier:

carriers	doping	total barrier	per carrier
3	43%	0.034 Ha	0.0113 Ha = 0.31 eV
2	29%	0.080 Ha	0.0400 Ha = 1.09 eV

The cost per carrier therefore *rises* as the carriers are thinned, by a factor of 3.5 for a reduction of 1.5 in density, and the total is not the sum of independent single-carrier costs. *The figure of 0.31 eV cannot be quoted as an activation energy*, and the withdrawal of the negative conclusion announced above is itself withdrawn.

The direction is what matters. A real semiconductor is doped at 10^{-6} to 10^{-3}, far more dilute than either row, so the relevant limit is the stiff end of this trend and not the soft one. Both figures were obtained at concentrations that are metallic rather than semiconducting.

The trend is nonetheless recognisable, and may be the framework behaving correctly. Dense donors have overlapping potentials which screen one another and flatten the landscape a carrier moves on; dilute donors leave each carrier in an isolated well it must climb out of. That is the impurity-band picture, and the crossover between the two regimes is the Mott criterion for doped semiconductors. What the measurements then say is that the framework sits on the wrong side of that crossover for real materials: it conducts when the carriers are dense enough to be nearly metallic, and localises them when they are not.

Three cautions belong with it, and none is small. The constraint is a modelling choice — these are the energies of a system in which carriers are forbidden the core — and it is exactly the choice that produces the result. The comparison above spans two changes at once, the constraint and the change from ring to straight chain, so the factor of sixteen cannot be attributed to either alone. And end effects are not excluded: at $d = 0.5$ the outermost carrier has one atom beyond it where the innermost has three, an asymmetry the mirror test cannot detect, since it forces both ends to behave alike and both are contaminated equally. A longer chain at the same doping would settle it, and has not been run.

Every barrier here is a rigid-lattice upper bound. One qualification belongs with all of the above, and it is not small. The nuclei are held fixed throughout, so what is computed is the cost of moving a carrier through a *frozen* lattice. A real hopping carrier is a polaron: it distorts the lattice around itself and carries that distortion along. Because the frozen geometry is the one appropriate to the pocket, the saddle is evaluated in a lattice optimised for somewhere else and is penalised accordingly, so relaxation lowers the saddle more than the pocket and *the true barrier is lower than the figures given.* How much lower is not known.

The attempt to measure it is reported as null rather than omitted. With the carrier held and the lattice released under damped dynamics, the energies came out 1.1–1.8 Ha *above* their frozen values and scattered by 0.09 Ha between repeats of the same configuration. Freeing degrees of freedom cannot raise a minimum, so those runs are not relaxed minima: with thirty electronic steps between nuclear moves and a convergence threshold ten times looser than that used elsewhere, the electronic state never overtakes the moving nuclei. The measurement wants the nuclei moved far more slowly, or a two-stage procedure in which the relaxed geometry is frozen and the electrons re-converged on it. Until then the polaron correction is unknown, and the conclusion that the framework does not conduct rests on upper bounds — which is the right direction for that conclusion to be wrong in.

What the frozen numbers do have is agreement across independent constructions. The interstitial barrier of Table 1 at $a = 2.5\,a_0$, 0.200 Ha, was reproduced to the last digit on a separate page written for the polaron test (-16.253 and -16.053 against -16.254 and -16.053), and the ring of §3 gives 0.180 Ha at $a = 2.2\,a_0$ with a carrier that localises unaided rather than by construction. Three figures, two geometries, one order of magnitude: about five electron-volts.

What is actually transported. The carrier in this framework is not an electron. An electron here is not a particle but a unit of charge density on a domain, and what exists is the *partition* of space; dynamics is the evolution of that partition, and transport is the partition pattern travelling.

What moves along the chain is the **free boundary** — a locus where two densities balance — and nothing material traverses it at all.

The quantitative form of this is the useful one. As a kink passes, each electron advances by exactly one site spacing and then stops, while the kink advances the entire length of the chain. The carrier's displacement and its constituents' displacement are decoupled, and their ratio is the chain length. This is the relation between a dislocation and the slip it produces: the dislocation crosses the crystal, no atom moves far. Charge is genuinely transported and real charge arrives at the far end; it is the *carrier* that is immaterial, in the sense that a wave is immaterial while the water is not.

That diagnosis also accounts for the failures recorded earlier in this study rather than excusing them. Each of the unsuccessful constructions imposed a carrier that had to *translate*, moving bodily through occupied space, and returned costs of tens of electron-volts or else collapsed. Those were the price of forbidding the only mechanism the framework has. The solver states as much directly: its free boundary evolves by the density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, flipping ownership cell by cell. Boundary motion is the only motion it performs. The corona result is the same statement seen from the favourable side — an added electron spread into a uniform shell at no cost, which is a boundary redistributing freely.

This raises the stake of the kink measurement. If transport works here it works *because* of the free boundary, which is the single feature separating this framework from the standard theory; and if the boundary cannot carry charge cheaply, the difficulty is not in a detail of the construction but in the framework's defining object.

Two results that may fit together as a semiconductor. Set that beside the corona result: a *single* extra electron, beyond a neutral complement of six atoms, delocalises and slides freely, while a full complement of six against bare kernels is pinned.

the filled electron ring (n electrons, n ions)	pinned, 323 meV — does not slide
one extra electron beyond it	delocalised, flat — no resistance

That reads as a semiconductor rather than a metal — a filled complement that does not conduct, an added carrier that does — and it would settle what the framework's difficulty with metals is: not that it cannot transport charge, since it transports surplus charge without any resistance at all, but that it may transport *only* surplus charge, while a metal conducts by the sea itself moving.

Why that reading is not yet established. The two runs differ in more than the filling, and the difference is not incidental. In the corona run the inner ring carries *neutral atoms*, each already holding its own electron, and the extra charge floats outside a screened, nearly field-free ring. In the sliding run the inner ring carries *bare* kernels, and the electrons under test are themselves the screening, sitting in the unscreened ion field. Flat in one case and corrugated in the other is therefore as well explained by screening as by filling.

In a real solid the two are the same statement: the filled valence electrons screen the cores, which is precisely why an added carrier moves in a flat potential. But that identification is imported rather than demonstrated here, and separating the two requires runs that differ in one respect at a time — the same bare-kernel ring at fixed geometry, with the electron count changed at fixed total charge (registry only, via fractional kernels) and with it changed at fixed kernel charge (filling as well). Until then the pinning number stands and the semiconductor reading is a conjecture about what produces it.

Two further qualifications belong with the number. Commensurate registry, n electrons on n ions, is the most strongly pinned case; a metal's sea is incommensurate with its lattice, which is part of why it slides, and whether the framework reproduces that distinction is not tested here. And the pinning reported is a static barrier, with no thermal or quantum reduction applied.

The barrier, which is a static quantity. Transport by hopping is governed by a barrier, and a barrier is an energy difference, which the static solver does compute. Placing an excess electron as a domain of its own at successive positions and relaxing each to convergence traces the surface the carrier moves on — the standard route to a small-polaron barrier.

A linear chain answers the wrong question: four hydrogens in a line give 1.16 Ha = 31.5 eV, but that is the cost of pushing the carrier *through* an ion, and stepping $1\,a_0$ aside makes it marginally worse rather than better, because in a wire there is no route from one interstitial pocket to the next except through an atom. A lattice has one. In a 4×4 square lattice the carrier passes from the centre of one square through the *midpoint of a bond* — flanked by two atoms at half a spacing each, never at zero distance from any — to the centre of the next.

lattice spacing	pocket (Ha)	saddle (Ha)	barrier
$a = 2.5\,a_0$	−16.254	−16.053	0.200 ± 0.003 Ha = 5.46 eV
$a = 1.4\,a_0$	−11.824	−11.402	0.421 ± 0.015 Ha = 11.5 eV

Table 1: Barrier for a carrier moving between interstitial sites of a hydrogen lattice. Uncertainties are from the mirror-symmetry check, the two bond midpoints either side of a pocket being equivalent by construction.

The lattice route is real: passing between two atoms costs 5.8 times less than passing through one, so the chain's figure was an artefact of dimensionality. But the barrier that remains is 5.5 eV, some $200\,k_BT$, and the carrier is confined to its pocket. At $a = 2.5\,a_0$ that may be right — nearly twice the H_2 bond length, where stretched hydrogen is a Mott insulator in nature too.

The density trend, and why the barrier scans mislead. Compressing the lattice did not close the barrier in those scans; it doubled it. But the scans share a defect that only became clear when the question was posed differently: they impose a *localised* carrier and measure what it costs to move. If the framework has no localised carrier, that cost is the price of maintaining an artificial constraint, not of transport.

A repair that is not available, and one that is. An obvious response to the barrier is to let valence electrons share a territory, which removes it by construction. Equation (3) rules that out:

sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude.

The interface is a surface tension, and that is why one number governs both problems. The parameter β has a physical identity that makes the conflict above intelligible rather than coincidental. Adding a surface energy $\frac{\beta}{2}\oint_{\partial\Omega}\psi^2\,dS$ to the localisation term and varying gives $\partial_n\psi+\beta\psi=0$ as the *natural* boundary condition — the Robin condition exactly. So β is a **surface tension of the electron–electron interface**, and $C=0$ says that this surface tension vanishes. A surface tension costs energy both to *have* and to *move*: the first is confinement energy, hence degeneracy pressure, and the second is resistance to transport. The two faces of β are therefore one quantity, which is why no value of it supplies the first without incurring the second. The hydrodynamic form of the framework, in which this identification is derived along with the observation that the localisation term is the von Weizsäcker functional and its variation is Bohm's quantum potential, is treated separately [3].

What may be relaxed instead is the requirement that each domain carry *exactly* one unit of charge. Suppose a domain may hold two — the excess electron is then not a new region competing for space, but additional occupation of a region that already exists. Two things follow immediately, and neither depends on any coefficient. First, by translational symmetry the energy is the *same* wherever the doubly-occupied site sits, so there is no barrier between sites at all. Second, electrostatics favours spreading the excess: with the interfaces free, the kinetic energy is unchanged at any occupation, so only the Coulomb terms act, and they are lowered monotonically as the excess is shared out. A localised double occupancy is therefore not merely mobile but unstable against delocalising.

The cost of forming it is the on-site repulsion, of order 10 eV for hydrogen-like sites — the Hubbard U, here *computed* from the geometry rather than fitted.

What the framework then is. Non-overlapping sites, occupation 0, 1 or 2, an on-site repulsion U, and transfer between neighbours: that is the **Hubbard model**. Which locates RealQM exactly. As formulated — one unit of charge per domain, always — it is the Hubbard model in the limit $U\to\infty$, where double occupancy is forbidden outright and every site is singly occupied. That limit is a Mott insulator at every density, which is precisely what Table 1 found by direct computation, and it explains the density trend as well: no finite U can be beaten by a bandwidth that does not exist.

The repair is correspondingly minimal. Keep the partition, keep the geometry, keep the Coulomb energetics, and allow occupation to take the value two at a computed cost. The framework then sits where the standard treatment of strongly correlated matter sits — the Hubbard model is the working description of Mott insulators, of the cuprates, and of the materials for which band theory fails — with the difference that its U and its lattice are derived rather than supplied.

We do not carry this out here. What is established is where the obstruction lies: not in the interface condition, not in the geometry, but in the rule of one unit of charge per domain. That rule is also the source of one of the framework's results, charge quantization as a consequence of indivisibility rather than an assumption. The tension is worth stating plainly: *the postulate that* *explains why charge comes in units is the postulate that forbids a metal.* Matter has it both ways, quantizing the particles while leaving the amplitude on each site continuous, and a partition of real densities has no evident means to do the same.

Where the boundary of the scope actually lies. It is fair to ask whether all conduction might be hopping, band transport being a formalism rather than an observable — which, if true, would remove the scope restriction of §2 altogether. It is not true, and the argument that settles it is the same one that fixes the boundary. The idea is not idle: hopping is the correct description of amorphous and organic semiconductors, of doped semiconductors and polaronic oxides, and in "bad metals" at high temperature the mean free path falls to the lattice spacing and the band picture fails there too. What settles it is the *low*-temperature behaviour of a pure metal. Hopping is thermally activated, so its conductivity vanishes as $T \to 0$; the resistivity of pure copper instead falls from $\approx 1.7\,\mu\Omega\,\text{cm}$ at 300 K to below $0.002\,\mu\Omega\,\text{cm}$ at 4 K, a ratio above 10^3 in good samples, with a mean free path reaching millimetres — 10^7 lattice spacings, which is not a sequence of jumps between neighbours. A hopping framework predicts that copper becomes an insulator on cooling, and the measurement runs the other way by three orders of magnitude. The Wiedemann–Franz ratio, fixed at $L_0 = \pi^2 k_B^2/3e^2$ by carriers that convey heat and charge alike, is a second such constraint.

The regimes are also the wrong way round for a framework attempting metals: hopping describes matter best at high temperature, and it is at low temperature, where band behaviour is cleanest, that it would be needed to hold. *That is the boundary, and it is a clean one.* Band transport in a pure metal is out of reach and is not claimed; activated transport by localised carriers is the regime the construction describes, and there the same features are the correct physics rather than a defect.

4 Metals: a Wigner–Seitz problem the framework already agrees with

Take a metal: a lattice of ions, one valence electron each. RealQM assigns each electron its own domain; by translational symmetry those domains are the Wigner–Seitz cells, one ion at the centre of each; and at a cell face, neighbouring cells being identical, the normal derivative of the density vanishes.

That is exactly the **Wigner–Seitz construction**, used for alkali metals since 1933. The framework does not improvise in condensed matter — it arrives already agreeing with an established method, and differs from it in one term only. Standard theory solves the same cell problem for the band bottom and then *adds* the mean kinetic energy of filling the band, $\frac{3}{5}E_F$, together with exchange and correlation, all consequences of the exclusion principle. RealQM keeps the electrostatics, takes its kinetic energy from the cell state itself, and adds nothing: non-overlap is supposed to do that work.

For each alkali metal, with an Ashcroft empty-core pseudopotential of radius r_c, electrostatics $-0.9/r_s + 1.5\,r_c^2/r_s^3$ for a point ion in a neutral sphere, exchange $-0.458/r_s$ and Wigner correlation, minimising the energy per electron over r_s gives Table 2. Each metal contributes one parameter, its core radius, and two measured numbers; across the series the equilibrium radius varies by 70% and the bulk modulus by a factor of six, so tracking the trend is a stronger statement than matching any single case.

		r_s (a_0)			B (GPa)		
	r_c	RealQM	standard	measured	RealQM	standard	measured
Li	1.06	2.40	2.86	3.25	49.6	23.9	11.6
Na	1.67	3.77	3.87	3.93	9.1	8.0	6.3
K	2.14	4.82	4.67	4.86	3.6	4.0	3.1
Rb	2.31	5.20	4.95	5.20	2.7	3.2	2.5
Cs	2.50	5.62	5.28	5.62	2.0	2.5	2.0
mean abs. error		6.2%	5.6%		79%	44%	

Table 2: The alkali series. On the equilibrium radius RealQM tracks the whole trend and is comparable to the standard treatment; on the bulk modulus, a second derivative, both are poor and RealQM is the worse.

The first thing to say about this table is that it does not discriminate. Repeating the calculation with the electronic kinetic energy set to *zero* — valence electrons sharing one territory, for which a uniform density has $\nabla\psi = 0$ exactly — moves the equilibrium radii by about one per cent (Na 3.73 against 3.77, Cs 5.59 against 5.62) and gives a mean error of 7.0%. Pure electrostatics, with no quantum treatment of the electrons at all, reproduces the alkali series about as well as either column above.

That is a property of simple metals rather than a defect of the calculation: the equilibrium volume is set by core repulsion against the Madelung attraction, and the electronic terms are small or cancel. The standard column reaches it by cancelling $\frac{3}{5}E_F$ against exchange; RealQM reaches it with both absent. *The lattice scale of an alkali metal is therefore not a test of the quantum treatment*, and the agreement in Table 2 should not be read as one. What it does show is that nothing in the framework's treatment of the electrons *spoils* a result electrostatics already gives — a weaker claim, and the accurate one.

The bulk modulus does discriminate, being a second derivative, and there the framework is the worse of the two: 79% mean error against 44%. Lithium is the outlier throughout, the empty-core description being known to serve it badly.

The mechanism of the agreement should be stated plainly rather than claimed as a triumph. At $r_s = 3.93$ in sodium the cell state carries $\langle T \rangle = 0.004$ Ha against a band-filling term $\frac{3}{5}E_F = 0.0715$ Ha, so RealQM is *not* reproducing degeneracy pressure. It arrives at a comparable answer because it also omits exchange, -0.117 Ha, a term of opposite sign and comparable size, so that their effects on the position of the minimum largely cancel. That cancellation is the framework's own design claim — that non-overlap does the work exchange does — and across five metals it holds for the radius and fails for the curvature, which is what one would expect of a cancellation that is close but not exact.

5 The same chain, computed the ordinary way

Everything above is internal to the framework. It is worth putting the same system beside the standard treatment, on quantities both can produce, and the result is a check on the scope claim of §2 rather than a contest.

Kohn–Sham DFT, PBE/6-31G, all electrons, a straight chain of lithium atoms at the spacings used above, with the geometry held uniform so that the comparison is like for like — a Peierls distortion is not permitted on either side:

	RealQM	PBE/6-31G	experiment
equilibrium spacing (a_0)	4.5	6.0	5.7 (bcc n.n.)
verdict at $a = 6.0$	pinned, 184 meV	metallic	lithium conducts
verdict at its own equilibrium	depinned, ≈ 0	metallic	—
cohesive energy (eV/atom)	~ 1.7	0.486	1.63 (bulk, 8 n.n.)

And it is not one functional's answer. Density-functional theory is not the Schrödinger equation: it replaces the interacting problem with fictitious non-interacting electrons in an effective potential, exact in principle by Hohenberg–Kohn but resting in practice on a *constructed* exchange–correlation functional and a finite basis — two uncontrolled approximations. The comparison above is therefore model against model, not model against exact, and would normally be qualified accordingly. On the quantity at issue it need not be. Repeating the scan with Hartree–Fock, which carries no correlation whatever, and with MP2, which carries it perturbatively, moves the total energy by some 0.3 Ha between methods and leaves the equilibrium spacing at 6.0 a_0 in all three:

a (a_0)	HF	PBE	MP2
4.5	-51.97126	-52.29669	-52.02861
5.0	-52.01844	-52.33386	-52.06750
6.0	$\mathbf{-52.04726}$	$\mathbf{-52.34332}$	$\mathbf{-52.08273}$
7.0	-52.04089	-52.31653	-52.06476

The twenty per cent is thus a disagreement with standard quantum chemistry rather than with a choice of functional.

But the objection cuts the other way, and harder. If one may choose the approximation to suit the problem then one will always win, and that is not a comparison. Applied evenly, the charge falls on *this* side of the table. The standard column used a non-empirical functional, a general-purpose basis, and three treatments of correlation that were not selected after the fact — but the RealQM column carries a free parameter, the pseudopotential radius r_c, and it was set to 1.6 a_0 by choice rather than by any independent criterion. At $r_c = 2.4$ the equilibrium spacing of the same chain moves out beyond 6.0 a_0, which *agrees* with the standard result. The twenty per cent is therefore a statement about $r_c = 1.6$ and not about the framework, and the framework could have been made to agree as easily as to disagree.

A comparison is only worth making if the parameters are fixed by something other than the quantity being compared. The proper protocol is to calibrate r_c on the *isolated* lithium atom — requiring the single pseudo-atom to reproduce the measured first ionisation energy, 0.198 Ha — and then to compute the chain with that value and no further adjustment. The chain is then a prediction rather than a fit, and stands on the same footing as the standard column, in which nothing was tuned to lithium chains. Until that is done the table above should be read as establishing the *method-independence of the standard answer* and nothing about the size of any discrepancy.
The metallic verdict is not read off a single gap, which any finite chain has, but off its scaling: 0.432, 0.346, 0.291, 0.247, 0.173 eV at $n = 5, 7, 9, 11, 13$, with $n \times$ gap constant near 2.4 eV, so the gap closes as $1/n$ and is finite-size rather than a band gap.

That verdict is weaker than it looks, and in a direction that matters. What was compared are Kohn–Sham eigenvalue differences, and Kohn–Sham eigenvalues are not excitation energies. They belong to a fictitious system of *non-interacting* particles introduced to reproduce a density; the true gap differs from theirs by the derivative discontinuity of the exchange–correlation functional, which PBE does not possess at all. Approximate functionals are known to underestimate gaps for exactly this reason, often by a factor of two. A small Kohn–Sham gap is therefore what the method returns whether or not a gap is present.

The bias has a direction, and it is the one at issue here. Approximate functionals carry a self-interaction error — an electron repelled in part by its own density — whose consequence is **delocalisation error**: densities spread too far, localised solutions are penalised, and metallic answers are favoured. So the standard column calling this chain metallic is that treatment erring in its characteristic direction, not a neutral measurement. Against it, the framework of this article over-localises, by construction, being $U \to \infty$. *Both are biased and in opposite senses*, and lithium is simply a case where the standard bias does no damage, because the chain really is metallic. On a system where it does damage — an oxide, a doped Mott insulator, anything whose gap is on-site repulsion — the delocalisation error is the well-known failure and the localisation is the correct instinct.

Two further asymmetries belong in the record. Density-functional theory admits no convergence path: wavefunction methods form a hierarchy, Hartree–Fock through MP2, CCSD, CCSD(T) to full configuration interaction, which converges to the exact answer within a basis, whereas no procedure refines PBE towards the true functional. And the quantities are not of the same kind. What is computed here are total energies of *specified configurations* — differences between states of one system, which are observables — where the gap above is an eigenvalue difference in an auxiliary system that was never claimed to have physical states. Where this framework is less accurate it is at least computing something that means what it says.

What this establishes, and what it does not. On this system the standard treatment is better and cheaply so: the equilibrium spacing within a few per cent of the measured nearest-neighbour distance, the right conduction verdict, a sensible cohesive energy, in seconds. RealQM places the atoms some twenty per cent too close, and at the spacing lithium actually adopts it predicts a pinning barrier of 184 meV where the metal has none.

That is not a surprise and should not be read as one, because *lithium is outside the scope this article claims.* §2 excludes metals in kind and not by degree: extended Bloch states, band transport, a Fermi surface, none of which the construction possesses. A simple metal is the standard treatment's best case and this framework's declared worst, so the table is a scope check and not a competition.

As a scope check it passes, in the specific sense that matters: the framework errs by *over-localising*, in bond length and in pinning alike, which is what $U \to \infty$ implies and therefore the direction the identification predicts. It fails as it was said it would, rather than arbitrarily.

Where the comparison would actually bite. The corollary is uncomfortable and is recorded plainly: nothing computed here is a quantity the standard treatment does not already supply more accurately. The case for the construction rests instead on the real-space mechanism — a registry, a sliding partition, a carrier that is a boundary rather than a particle — being a more transparent account of activated hopping than a model Hamiltonian with fitted parameters. That claim is not tested by lithium. It would be tested on the systems the scope table names, where on-site repulsion rather than band structure sets the gap and where density-functional theory is itself in difficulty: a doped Mott insulator, an organic semiconductor, a polaronic oxide. Those comparisons are not made here.

6 The interface coefficient, and the one number that is wrong

Equation (2) was introduced above for what it gives conduction. It has a second face, and this section is that face: the same vanishing interface cost, read as a statement about the *equation of state*, where it is a defect rather than a mechanism, and where — this being the useful part — it turns out to be a single misvalued number rather than a structural incapacity.

Partitioning a gas of density n into non-overlapping cells of size $a = n^{-1/3}$ gives a localisation cost per cell scaling as a^{-2}, hence a kinetic energy density $t(n) = C\, n^{5/3}$ — the Thomas–Fermi form, obtained from the geometry of a partition rather than from Fermi statistics. Linearising (1) about the uniform state and substituting into Poisson's equation gives Yukawa screening with $k^2 = 18\pi n^{1/3}/5C$, and setting C to the Thomas–Fermi coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces k_{TF} exactly at every density. *The form extends without adjustment.*

That last statement should be read for exactly what it is worth, which is less than it sounds. Thomas–Fermi screening is itself derived from the Thomas–Fermi kinetic energy by precisely these steps — $\mu = \partial t/\partial n$, then $\partial n/\partial \mu$, then Poisson — so recovering k_{TF} from C_{TF} is an algebraic identity and not a second, independent agreement. What it establishes is that the framework's linear response is the correct derivative of its own kinetic energy, and what it *provides* is a translation of the abstract coefficient C into an observable length. That translation is the reason the screening relation is worth carrying: it is how the value of C becomes a statement about matter.

The coefficient does not. By (2) the flat trial function is admissible, so $C = 0$ exactly, and neither allowing the interfaces to move nor requiring domains to be connected and singly charged alters it.

The coefficient is a boundary condition, and its value is known. What sets C is the condition imposed where one domain meets the next. Solving the cell problem under a Robin condition $\partial_n\psi + \beta\psi = 0$ at the face, and fitting β to the electron gas, gives:

	C	C/C_{TF}
Neumann, $\beta a = 0$ (the present choice)	0	0
Robin, $\beta a = 1.146$	2.871	**1.00**
Dirichlet, $\beta a \to \infty$	14.804	5.16

All three carry the same, correct, $n^{5/3}$ scaling; *they differ by a pure number*, and the framework has been using the one value of that number for which confinement is free. This is why the Thomas–Fermi comparison is made first and not last: it converts the finding from "the construction has no degeneracy pressure" into "the construction has one interface parameter set to zero," and supplies the value it would need. The structure is not in question. One number is.

Why Neumann is free, and why it went unnoticed. A constant satisfies Neumann conditions on every face, so the ground state of $-\frac{1}{2}\nabla^2$ in a Neumann cell has energy *exactly* zero at any cell size. Confinement costs nothing: halve the cell and the energy is unchanged. That is a failure of the uncertainty principle at precisely the place where electron meets electron — and degeneracy pressure is nothing other than the energy cost of confining fermions. The substitute for exclusion reproduces its bookkeeping and none of its energy.

It went unnoticed because every previous test of the framework had an attractor in it. Where a nucleus shapes the density, $\nabla\psi \neq 0$ for reasons having nothing to do with the interface, and T comes out right; the faces are nearly irrelevant. Remove the attractor and the faces are all that is left.

The cost of the mechanism. Read as a statement about *structure*, the free interface is destructive. There is no degeneracy pressure; the screening length *vanishes*, since $k^2 = 18\pi n^{1/3}/5C$ diverges as $C \to 0$, so the framework does not fail to screen but screens perfectly at zero range, charge piling wherever the potential is low at no cost in kinetic energy; the plasmon is dispersionless; there is no bandwidth; and no mechanism remains by which a metal could differ from an insulator. Over-screening and the over-delocalisation measured in §3 are the same phenomenon.

Every one of those is the price of the mechanism §3 runs on. The defect that empties out degeneracy is the same one that lets charge flow, which is why the two halves of this paper cannot be repaired independently.

What is being matched, and what is not. Thomas–Fermi is a poor theory and no weight is placed on it here. It overbinds helium by about a third, has no shell structure and therefore no periodic table, and by Teller's theorem it binds *no molecule at all*: in Thomas–Fermi theory separated atoms are always lower in energy than any molecule. What is exact is narrower and is the only part used. The coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3}$ is the exact kinetic energy density of a *uniform* degenerate electron gas — a theorem about free fermions rather than a model, Thomas–Fermi theory being what results from applying that exact local statement to inhomogeneous atoms, which is where it fails. The comparison above uses the one rigorous number in it, in the one limit where it is rigorous.

That said, the two constructions stand in a suggestive relation, and it bears on how freely β may be chosen:

	uniform gas	molecular binding
Thomas–Fermi	exact	none at all (Teller)
RealQM as formulated	$C = 0$	its entire record of results

They are mirror images: each is empty exactly where the other is exact. And βa is a dial running between them, 0 at the RealQM end and 1.146 at the Thomas–Fermi end. The question of whether the electron gas's value is tolerable for atoms and molecules is therefore not a technical check but the substantive one — *does moving towards the Thomas–Fermi coefficient destroy binding, as it does in Thomas–Fermi theory?* Teller's theorem is a standing warning that the far end of the dial is hostile to molecules.

One dial, pulling both ways. It should be said at once that β cannot be tuned to satisfy both readings independently, because a single parameter governs them. Raising βa from zero towards 1.146 buys degeneracy pressure, a finite rather than vanishing screening length, and a dispersive plasmon — and in the same movement makes the interface cost energy to move, which is resistance. The value that repairs the structure is the value that impedes the transport. Whether some intermediate setting serves both, or whether the two demands are simply in conflict, is not settled here, and it is the sharpest form of the question this paper leaves open. Whether the single value the electron gas requires is even tolerable for atoms and molecules is a second such question; the attempt is reported in the numerical note.

The dial, measured: the far end unbinds H_2. That question can be answered rather than left open, and the answer is unfavourable. Adding the surface term to a two-domain calculation and scanning β at fixed separation gives the binding energy directly. For H_2 at $R = 1.6$ a_0 with the interface held at the midplane, against $2E(\mathrm{H}) = -0.85190$ on the same grid:

β (a_0^{-1})	$E(H_2)$	E_{bind}
0	−1.02757	+0.17567
0.02	−1.01428	+0.16238
0.05	−0.99439	+0.14249
0.1	−0.96143	+0.10953
0.2	−0.89700	+0.04510
0.4	−0.77889	−0.07301
1.146	−0.50023	−0.35167

The response is linear at small β — $\Delta E_{\mathrm{bind}} = -0.665\,\beta$ to four figures for $\beta \leq 0.05$ — and the bond is gone by $\beta \approx 0.28$. To compare against the electron gas, which fixes βa rather than β: two electrons at bonding density give $a = n^{-1/3} \approx 2.55$ a$_0$, so binding survives only to $\beta a \approx 0.7$, while the degenerate gas requires $\beta a = 1.146$. *The two demands are in conflict, not merely in tension*, and Teller's warning is confirmed inside this framework rather than inherited from Thomas–Fermi theory. The intermediate setting that serves both does not exist for this molecule.

But β remains available as a regulariser. The linearity is worth separating from the physics, because it licenses a use of β that carries no commitment to any value of it. At $\beta = 0$ a free interface has no restoring force, and the free-boundary energies scatter: repeated runs of H_2 at $R = 6$ under identical settings spread over 0.3 Ha, while $E(\mathrm{H})$, which has no interface, held to 10^{-3}. A small β suppresses that scatter, and since its cost is linear the physical answer is recovered by fitting two small values and extrapolating to $\beta \to 0$ — 0.02 and 0.05 cost 0.36 and 0.90 eV of binding respectively, and the intercept removes both. Used this way β is a numerical parameter with a limit, not a material constant, and the unbinding above is irrelevant to it. The two roles should not be confused: the dial is a physical claim, the regulariser is not.

What the partition supplies, and why it cannot be given up. Since (2) arises from the freedom of the interface, an obvious thought is to drop the partition and let the electrons share one territory. The cost of that is fixed by a scaling argument, and it is prohibitive. Confining N unit charges to one shared region of volume $V = N/n$ gives a kinetic energy density

$$\frac{T}{V} = \frac{3\pi^2}{2}\, n^{5/3} N^{-2/3}, \tag{3}$$

against the $n^{5/3}$ of the partition with no N in it at all. The partition is what makes the kinetic energy *extensive*: each electron is localised to a cell whose size is set by the density, not by the size of the sample. Sharing makes it depend on how large a piece of metal one happens to be considering, and at $N \sim 10^{23}$ the density falls by $N^{-2/3} \sim 10^{-15}$ — fifteen orders of magnitude. *The partition must be kept*, whatever else is relaxed.

What (2) is, and is not, a statement about. It concerns a *uniform* density and does not transfer to a shaped one. Where a nucleus shapes the profiles — a two-domain shell on one kernel, for instance — both domains carry real kinetic energy, and the shell is stable under the density-balance rule, since compressing the inner domain raises its density at the interface and lowers the outer's, pushing the boundary back out. The vanishing is a property of the flat case alone.

That also settles how much weight the uniform electron gas should carry. It is electrons plus a *smeared, inert, non-responding* positive background — a construction belonging to a tradition in which exchange–correlation energies are fitted to that very system. RealQM contains no inert backgrounds anywhere, so asking it to reproduce jellium asks for a system whose central ingredient it does not have, and $C = 0$ diagnoses that mismatch rather than recording a failed test. Consistently, the pathology disappears as soon as real ions are present: Table 2 is the same framework on the same physics, with the background replaced by the charges matter is actually made of.

One physical case does approach the uniform limit closely: degenerate matter in which the ion lattice is a small perturbation, as in a white dwarf interior, where the measured mass–radius relation depends on precisely the pressure the uniform limit does not supply. That is the one place the result might have observable consequences, and it is untested here.

7 The high-temperature limit

Conduction above is the elementary case and the hard one. A hot classical plasma is the extreme case and the *easier* one, because degeneracy is not required of it: the physics there is Coulomb interaction and kinetics, which is what this framework is made of, and nothing in it leans on the pressure that (2) removes. The order of difficulty is the reverse of the order of familiarity.

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{4}$$

free of temperature and of any statistical assumption. RealQM reproduces this exactly, for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on.

The same structure suggests what the framework should do about **Landau damping**. RealQM is not a single fluid: N electrons are N domains, each carrying its own current and drift velocity, which is a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, collisionless damping should therefore emerge for the right reason rather than by an inserted term. This has not been computed.

8 Out of scope: the Fermi surface, and what that costs

The framework has no bands and no E_F, and there are measurements bearing on this directly. They are made on real two-component metals, not on any idealisation, so they cannot be set aside by choosing the object of study more carefully:

- **de Haas–van Alphen oscillations** map the Fermi surface of copper and aluminium directly; its existence and shape are not in doubt;
- the **electronic specific heat** is linear in temperature, with a coefficient fixed by the density of states at E_F;
- the **plasmon dispersion** is positive and measured, while the compressibility term that produces it, $\omega^2 = \omega_p^2 + \frac{3}{5}k^2 v_F^2$, is absent here.

No account of these is available in the framework, and none is attempted. Within the scope of §2 that is not a debt: every measurement in the list is made on a *degenerate* electron gas, and a dilute carrier gas has no Fermi surface to map, no E_F to fix a density of states at, and no v_F to disperse a plasmon with. A theory of localised carriers is not obliged to supply them.

The price should nonetheless be stated plainly rather than defined away, since it is what the scope restriction buys. *Metals are out of reach* — not approximately, but in kind. It is not disposed of by the equation of state, since the lattice scale of a simple metal turns out to be electrostatic and to test no quantum treatment at all (§4). It is not disposed of by the conduction repair, since a shared valence territory conducts while still having no band structure and no E_F. And it is not an artefact of any idealisation, being measured on ordinary metals. A set of Fermi-surface phenomena simply has no representation in a theory built from densities on domains, however those domains are constrained. Anyone wanting metals from this construction must supply extended states, and nothing in this article suggests how.

A note on what was tested numerically

Equation (4), the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. Table 2 is a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *The Hydrodynamic Form of RealQM: Madelung Hydrostatics, a Surface Tension, and the Missing Half*, in [2].

[4] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].